Recording and Performance Evaluation

Importance of Records

- Livestock rearing has changed **from backyard or subsistence type to an organized and commercial type** very rapidly over the past few decades
  - Two major consequences:
    - Establishment of larger livestock farms involving higher investment
    - Difficulty to maintain information on each individual animal in one's own mind as it was done in the past
  - Demands maintaining the various types of information on production, reproduction, stock, human labor, etc. in written forms, i.e., records

Performance recording Cont . . .

- **Records help to:**
  - Evaluate performance of animals and formulate **appropriate selection and breeding** strategies
  - Design breeding **calendars** for individual breeding female, i.e., cows, ewes, does, sows, etc.
  - Accurately **inform the farmer** about what has actually happened in the working of his farm
  - Records aid in keeping **systematic checks** on daily farm routines
  - Enable to keep finger on the **pulse of daily happenings** of the farm

Essential features of recording…

- A system of identification so that each animal can be properly and permanently identified
  - Simplicity to register information
  - Should require minimum recording time
  - Should contain all necessary information

Essential features of recording…

- Accuracy
  - Data should have authenticity/reality
  - Data should be up-to-date
  - Recording must be made on the day of occurrence
  - Accessible: info needed must be easily accessed
  - Avoid unnecessary duplication

Benefit of well-kept records

- Pedigree records can give some **indication of the genetic potential** of animals
  - Performance records complemented with pedigree records enable to accurately **evaluate the breeding worth** of each animal
  - Enable assessment of **rate of success** after executing a designed breeding program for some time
  - Facilitate **identification of mistakes** of management practices like feeding, housing, milking, etc. and which aspect should be improved

Types of Records

- Identification record
- Pedigree record
- Performance records:

– Female performance (daily milk yield, lactation milk yield, number weaned, weight weaned, etc.)

– Growth and weight records (birth weight, weaning weight, yearling weight…)

– Breeding records (heat detection, mating/insemination, sire, expected & actual due date, …)

– Calving/lambing/kidding… record

- Health (vaccination, disease, trt, post-mortem …)
- Financial records
- Feeding records

Animal identification

- All animals should be identified and registered
  - Each animal's ID should be:
  - Unique to that animal
  - Given to the animal at birth
  - Never be used again for any other animal
  - Be used throughout the life of the animal in the country of birth and also by all other countries
  - **NB**:– When implementing animal identification for the first time, assign the **smallest number to the oldest animal** in the herd/flock so that sires/dams logically appear before their offspring

Identification

Permanent

- Hot iron branding
- Tattooing
- Freeze branding
- Ear notching
- Microchip implants

Temporary

- Ear tags
- Electronic ear tags (RFIDs)
- Ear buttons
- Electronic collars
- Rumen boluses with microchips installed
- Neck chains
- Wing bands
- Ankle straps
- Marking paint
- Leg bands

Permanent identification types

Hot iron branding





Freeze branding







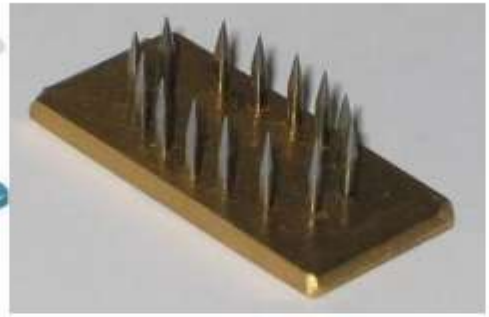
Microchip implant with applicator



Permanent identification…

Tattooing

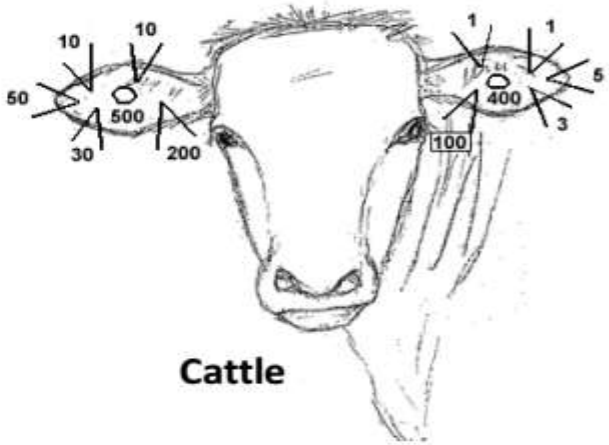




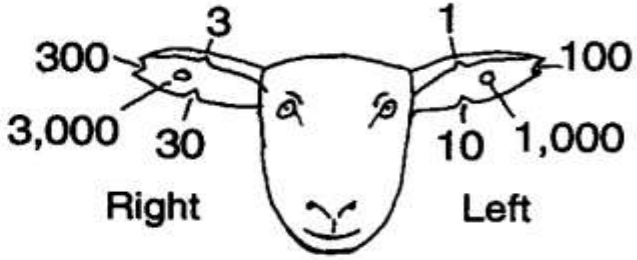




Ear notching



Cattle



Sheep

Temporary identification

Ear tags (metallic or plastic)









Neck chains



Rumen bolus & balling gun





Temporary identification…

Leg bands







Wing bands







Record formats

1. DAILY LIVESTOCK REGISTER

Date	Cows		Bulls	Calves		Heifers	Bullocks	Total Livestock	Milk yield			Additions during day		Deductions during day		Remarks
	Milk	Dry		Male	Female				AM	PM	Total	№	From where	№	How	

2. CATTLE BREEDING REGISTER

Ser №	Cow ID	Date of Calving	First Service			Second Service			Third Service			Fourth Service			1st PD (Date & Result)	2nd PD (Date & Result)	Expected Calving Date	Date on Which to be Dried off	Actual Calving Date	Weight of Calf	Sex of Calf	Number of Calf	Remark
			Date	Time	Bull	Date	Time	Bull	Date	Time	Bull	Date	Time	Bull									

Record formats…

3. CALVING REGISTER

Cow ID №	Due Date	Date of Calving	Sire	Calf №	Calf Sex	Birth Weight	Breed	Remarks

4. CALF REGISTER

Ser №	Birth Date	Numbering Date	Calf ID №	Dam	Sire	Sex	Birth Weight	Disposal		Remarks
								How	Date	

Record formats…

5. GROWTH RECORD OF YOUNG STOCK

Ser №	ID №	Birth Date	Birth Weight	Weight at Weeks												Weight at Months												Wt at P	Wt at S	Remarks
				1	2	3	4	5	6	7	8	9	10	11	12	4	5	6	8	10	12	14	16	18	20	22	24			

6. CATTLE HISTORY AND PEDIGREE SHEET

Animal ID №	Date of Birth or Date of Purchase & age at purchase	Sire №	Dam № ____	1st Lactation	2nd Lactation	3rd Lactation	4th Lactation	5th Lactation	PGD № ____	MGS № ____	MGD № ____
			Dam's MY								
			Dam's LL								
			Dam's DD								

Record formats…

7. REPRODUCTION AND PRODUCTION

Ser №	Service		Calving			Milk Yield								Remark
	Bull №	Date	Date	Sex of calf	Dam's Wt at calving	Total 305 Days	Lactation Length	Date Dried	Days Dry	Peak Yield	Date of Peak Yield	Reason for Incomplete Lactation	Service Period	Calving Interval	

8. DAILY MILK YIELD RECORD FOR THE MONTH OF _____

Ser №	ID №	Date of calving	Date																Total
			1st		2nd		3rd		4th		5th		…		30th		31st		
			AM	PM	AM	PM	AM	PM	AM	PM	AM	PM	AM	PM	AM	PM	AM	PM	

Record formats…

9. LACTATION MILK YIELD RECORD

Cow ID №/Name: _____
Breed: _____
Birth Date: _____

Dam's ID/Name: _____
Sire's ID/Name: _____
Farm: _____

Recording Year	Parity	Calving Date	Monthly Milk Yield												Total
			Jan.	Feb.	March	April	May	June	July	Aug.	Sept.	Oct.	Nov.	Dec.	

Record formats…

10. REPRODUCTION RECORD (ANIMAL CARD)

Cow ID **№**/Name: _____ Dam's ID/Name: _____

Breed: _____ Sire's ID/Name: _____

Birth Date: _____ Farm: _____

	Lactation Number								
	1	2	3	4	5	6	7	8	9
Breeding/Insemination Date									
Sire's ID №/Name									
Number of Services/Breeding per conception									
Date Expected									
Actual Calving Date									
Gestation Length (days)									
Calving Interval (days)									
Total Milk Yield (kg)									
Date Dried off									
Lactation Length (days)									
305-days Lactation Milk Yield (kg)									
Per cent Fat									
Calf ID №									
Sex of Calf									
Birth Weight of Calf									
Weaning Weight of Calf									
Remarks									

Record formats…

11. HERD HEALTH REGISTER

Date	ID №	When trouble started (history)	Symptoms	Diagnosis	Treatment	Result Cured/ Dead/ Sold	Name of Veterinarian	Cost of treatment	Detail of vaccination	Remarks

12. DAILY FEEDING REGISTER

Date	№	Concentrates (kg)			Green fodder (Qt)			Dry fodder (Qt)			Other feeds		
		Received	Issued	Balance	Received	Issued	Balance	Received	Issued	Balance	Received	Issued	Balance

Standardization of Records

- For accuracy in selection the record should reflect as precisely as possible the animal's genetic potential for the trait
  - However, several environmental factors obscure true genetic ability
  - Performance records should be adjusted for known environmental effects so that to do fair comparison of animals
  – Remove some of the environmental variation

Standardization of Records

- **Example performance recording (in dairy cows)**
  - Lactation length
  - Age/parity
  - Number of times milked daily
  - Length of the preceding dry period
  - Days open
  - Herd, season, year differences…
  - **Two methods are commonly used:**
  - Correction factors
  - The least squares procedure

Standardization of Records

- **Correction Factors**
  - In principle, correction factors are only valid for the population from which they are estimated
  - Estimates made within herds are, however, imprecise because of the small amounts of data, in addition to the heavy computational work involved
- In practice, correction factors are estimated from data of many herds in a region to be applied in all herds in that environment

Correction for environmental factors

- Effects of factors should be taken into account when comparing the performance of different animals

- Adjustment or correction factors should be applied before some data can be analyzed and comparisons made.

➢ **To have precise estimate of breeding value, data should be recorded and standardized for known environmental effects**

Month of calving: A cow calving in dry months may produce less milk than those calving in wet months

Month of calving	Correction factor
Jan.	0.95
Feb	0.98
Mar	0.99
Apr	1.1
May	1.2
Jun	1.3
Jul	1.1
Aug	1.05
Sep	1.05
Oct	1.1
Nov	1
Dec	0.99

For example

- Cow1 – give birth in Jan and produce 2000 kg milk yield
- Cow2 – give birth in Aug produce 1905 kg milky yield

✓**Before correction-** cow1 is superior than cow 2

- **Adjustment**

- For cow 1= 2000 kg x 0.95 = 1900 kg
- For cow2 = 1905 kg x 1.05 = 2000.25 kg

✓**After correction** cow2 is selected

Cont…

- **Frequency of milking:** Milking thrice will result in more milk yield (15 – 20 %) than twice and milking once will result in less milk (10 – 15 %) than twice milking.

Number of day milked 3x and then milked 2x	Correction factor equivalent to 2x milking
5-15	0.99
16-35	0.98
36-55	0.97
56-65	0.97

Correction factor for fat percent: Fat percentage is inversely proportion to milk yield.

If a cow produces 2000 kg milk in lactation with 6% fat, the FCM (Fat Corrected Milk) would be as fallows:

- 4% FCM = Actual lactation milk yield x actual fat% / 4% fat

= 2000 x 6/4

= 3000 kg milk

Correction for lactation length:

- Usually **305** days of lactation length is **standard**
- In tropics short lactation is common
- Cows drying before 200 days or yielding more than 400 days are excluded from the data for genetic analysis
- The following correction factor will be used if an animal has below 305 days lactation length.

Cont…

Days milked	Correction factor
95	2.85
96	2.61
97	1.77
215	1.32
245	1.18
275	1.08
305	1

Cont…

➢Correction for weight:

✓For beef

- For weaning weight
- AWWT = $\frac{\text{actual WWT- BWT}}{\text{age (days)}} \times 205\text{days} + \text{BWT}$

- AYWT - adjusted for 365 days

$$= \frac{\text{Actual final WT- actual WWT}}{\text{No. of days B/n weighting}} \times 160 + \text{AWWT}$$

✓For sheep and Goat

$$\text{AWWT} = \frac{\text{actual WWT} - \text{BWT}}{\text{actual age (days)}} \times 120 + \text{BWT}$$

Least Squares Procedures

- Is a form of **mathematical analysis** that finds the line of best fit for a dataset providing a visual demonstration of the relationship between the data
- Aims to create a **straight line that minimizes the sum of the squares of the errors**

  Requires **simultaneous consideration of all effects** and it gives **unbiased estimates**
  - It disentangles the different kinds of environmental effects
  - The system used is the method of fitting constants

Least Squares Procedures

- Before the data can be prepared for least squares analysis by computer, the various environmental factors to be considered should be included in the model
  - – Continuous variables, e.g. age at weaning, and
  - – discrete classes, e.g. sex
  - – Interactions may also be included but they increase the amount of computations
  - • On the basis of preliminary analysis, some of the classes can be omitted

Performance Evaluation

- Selection decision based on phenotypic value:

  P = A + NA + E, where

  - P = observed phenotypic value of an individual
  - A = additive genetic effect (variance)
  - NA = Non-additive genetic effects (variances due to dominance and epistatic gene action)
  - E = environmental effects (variance)
  - Which component would you select on?
    - A? NA? E? All?

Performance Evaluation

- **Additive genetic variance** is the component of variation that is transmitted to offspring
  - Additive genetic variance is also referred to as **breeding value**
  - Breeding value = value of genes to progeny
  - Genetic value = value of genes to self; includes all types of gene action (additive, dominance, and epistatic)
- **Additive genetic effect**: two or more genes source a single contribution to the final phenotype, or when alleles of a single gene (in heterozygotes) combine so that their combined effects equal the sum of their individual effects
- **Epistasis:** the phenomenon where the effect of one gene (locus) is dependent on the presence of one or more 'modifier genes

Neither NA nor E can be passed on to progeny

Breeding value (BV)

- The breeding value is part of the **genotypic value** that is transmitted to the offspring.
- The part of an individual genotypic value that is due to additive effect and therefore **transmittable** (true breeding value)
- **Genetic merit** of an animal for a given trait. The value of an animal as a parent.
- Additive Effect- Independent Gene Effect
- Additive Genetic Value = Breeding Value.

Breeding value (BV) continued

- If the breeder chooses individuals to be parents according to their phenotypic values,
  - ✓ *His success in changing the characteristics of the population depends on the degree of correspondence between phenotypic values and breeding values.*

Breeding value (BV) continued

- If all the variation is attributable to environment, selection of phenotypically superior individuals does not result in any alteration in the next generation.

✓**Desired characteristics of estimated breeding value**

- EBV should be an unbiased predictor of the true breeding value
- EBV should be as accurate as possible

Estimation of BVs continued

- In real life we observe phenotype (B) but want to estimate Additive genetic effect (A)
- Suppose that the phenotype of parent is +3 (expressed as a deviation from the population mean) for a trait with heritability of 0.33
  - Then, what is its BV (i.e. values of its genes to its progeny)?
  - The contribution of each effect is proportional to the variance explained by the effect
- ***VarP = VarG + VarE + Var(G×E),*** where $\sigma^2_G = \sigma^2_A + \sigma^2_D + \sigma^2_I$

$$\hat{A} = \frac{V_A}{V_P} P$$
Additive (BV)

$$\hat{D} = \frac{V_D}{V_P} P$$
Dominance

$$\hat{E} = \frac{V_E}{V_P} P$$
Environment

Estimation of BVs

- The prediction of **BVs** constitutes an integral part of most breeding programs for genetic improvement
- Crucial to the accurate prediction **BV** is the availability of records
- In a population, data available at the initial stages are usually on individual animals
- Later on offspring and other relatives. Thus, initially, the prediction of breeding values may be based on the records of individuals and few relatives.

Estimation of BVs: simple approach

$$\hat{A} = \frac{V_A}{V_P} P = h^2 P = EBV$$

$h^2 = VA/Vp$

- Example 1:
- Given:
  - P = +3.0 units (above the population mean)
  - $h^2 = 0.33$
- Estimated BV (EBV) = 3.0 * 0.33 = +1 units
- The breeding value of this parent is 1 unit superior to the herd/flock average

Using EBVs: Example 2

- Given:
  - Ram = 60 kg
  - Ewe = 50 kg
  - Average of flock = 45 kg
  - $h_2 = 0.25$
  - Ram EBV = P * h^2 = 15 * 0.25 = 3.75 kg
  - Ewe EBV = P * h^2 = 5 * 0.25 = 1.25 kg
  - Expected genetic value of progeny (on average):

  = (3.75 + 1.25)/2 = 2.5kg (above population average)

Genetic Evaluation with Different Sources of Records

1. The Basic Model

- The **phenotypic observation** on an animal is determined by environmental and genetic factors and may be defined by the following model:

Phenotypic observation = environmental effects+ genetic effects + residual effects

Or

$$y_{ij} = \mu_i + g_i + e_{ij}$$

Cont…

- Where; y_{ij} is the record j of the ith animal;
- μ_i refers to the identifiable non-random (fixed) environmental effects such as herd management, year of birth or sex of the ith animal;
- g_i is the sum of the additive (ga), dominance (gd) and epistatic (ge) genetic values of the genotype of animal i
- **eij** is the sum of random environmental effects affecting animal i

Cont…

- The additive genetic value in the **g** term above represents the average additive effects of genes an individual receives from both parents and is termed the ***breeding value*** (BV).
- Each parent contributes a **sample half** of its genes to its progeny
- The BV of the progeny therefore is the **sum of the transmitting abilities of both parents**

Cont…

- Since the **additive genetic value is a function of the genes transmitted** from parents to progeny, it is the only component that can be selected for & therefore the main component of interest

- In most cases, **dominance** and **epistasis**, which represent intra-locus and **inter-loci** interactions, respectively, are assumed to be of little significance and are included in the **eij** term of the model

- **eij** being the sum of the random environmental effects, dominance and epistatic genetic values.

Cont…

- The sampling nature of inheritance implies that each parent passes only a sample one-half of their genes to their progeny.

- There is, therefore, genetic variation between offspring of the same parents since all offspring do not receive exactly the same genes.

Cont…

- if a_i is the BV of animal i then

$$a_i = g_{ai} = \frac{1}{2}a_s + \frac{1}{2}a_d + m_i$$

- Where: a_s = BV of the sire

  a_d = BV of the dam,

  m_i = is the deviation of the BV of animal i from the average BV for both parents, that is, Mendelian sampling.

BV Prediction from the Animal's Own Performance

1) Single Record: **When one phenotypic record is the only available information on each animal, the estimated BV (EBV) for animal i can be calculated as:**

$$EBV = b\left(P - \overline{P}\right) \qquad Acc = \sqrt{b}$$

P - phenotype of the individual

$\overline{P}$ is the average phenotype of the herd (contemporary group)

b is a regression coefficient

In this case $b = h^2$.

Example 1: ***Given that the yearling weight of a heifer is 320 kg in a herd with a means of 250 kg, predict her BV & its accuracy if the h^2 of yearling weight is 0.45.***

$$EBV = h^2\left(P - \overline{P}\right) = .45\left(320 - 250\right) = 31.5\,kg$$

$$Acc = \sqrt{b} = \sqrt{.45} = 0.67 = 67\%$$

2. Repeated records

- When multiple measurements on the same trait, such as milk yield in successive lactations, are recorded on an animal its breeding value may be predicted from the mean of these records

.

When using repeated records, the formula for EBV & b are:

$$\hat{a}_i = b(\bar{y}_i - \mu)$$

$$b = \frac{nh^2}{[1+(n-1)R]}$$

Repeatability can be calculated

- Formula: **$R = (\sigma^2G + \sigma^2Ep)/ \sigma^2P$** i.e. ratio of genetic & permanent environmental variance to total phenotypic variance.

OR

$$R = \frac{\sigma^2A + \sigma^2D + \sigma^2I + \sigma^2Ep}{\sigma^2A + \sigma^2D + \sigma^2I + \sigma^2Ep + \sigma^2Et}$$

- Where R = Repeatability

σ^2G = Genetic variance,

σ^2P = Total phenotypic variance

σ^2A = Additive genetic variance,

σ^2D = Dominance variance

σ^2I = Epistatic variance

σ^2Ep = variance due to permanent environmental effect

σ^2Et = variance due to temporary environmental effect

Example 2: *Assume that a cow has a mean yield of 8000 kg of milk for its 1st & 2nd lactations. If the phenotypic STD & h^2 of milk yield in the 1st two lactations are 600kg & 0.30 respectively & the correlation b/n 1st & 2nd lactation yields is 0.5, predict the BV of the cow for milk yield in the 1st two lactations & its accuracy. Assume that the herd mean for 1st & 2nd lactations is 6000 kg.*

Solution: $n = 2$; $h^2 = 0.3$; $R = 0.5$;
Cow's mean for 1st & 2nd lactation = $\overline{Y}_i = 8000$
Herd's mean for 1st & 2nd lactation = $\mu = 6000$

$$b = \frac{nh^2}{1 + (n-1)R} = \frac{2x0.3}{1 + (2-1)0.5} = \frac{0.6}{1.5} = 0.4$$

$$\hat{a}_i = b(\bar{y}_i - \mu)$$

Therefore:

$$\hat{a}_{cow} = 0.4(8000 - 6000) = 800 \text{ kg}$$

and:

$$r_{a,\tilde{y}} = \sqrt{0.4} = 0.632$$

3 BV Prediction from Progeny Records

For traits where records can be obtained only on females, the prediction of BVs for sires is usually based on the mean of their progeny.

This is typical of the dairy cattle situation, where bulls are evaluated on the basis of their daughters.

Let $\bar{y}$ be the mean of single records of *n* progeny of sire *i* & assume that the progeny are only related through the sire (*paternal half-sib*).

Then the BV of the sire i is:

$$\hat{a}_i = b(\bar{y}_i - \mu)$$

Where

$$b = \frac{2n}{n + k}$$

with

$$k = \frac{4 - h^2}{h^2}$$

Where; n= number of progeny ; k is constant for any assumed heritability

The accuracy of the estimated BV is:

$$r_{a,\bar{y}} = \sqrt{\frac{n}{n + k}}$$

Example 3: ***Suppose the fat yield of 25 half-sib progeny of a bull averaged 250kg in the first lactation. Assuming a h^2 of 0.3 & a herd mean of 200 kg. Predict the BV of the bull for fat yield & its accuracy.***

Solution: $n = 25;\ \bar{y} = 250;\ h^2 = 0.3;\ \mu = 200$

$$k = \frac{4 - h^2}{h^2} = \frac{4 - 0.3}{0.3} = 12.3$$

$$b = \frac{2n}{n + k} = 1.34$$

$$\hat{a}_i = b(\bar{y}_i - \mu)$$

4 BV Prediction from Pedigree

When an animal has no record, its BV can be predicted from the evaluations of its sire (s) & dam (d). Each parent contributes half of its genes to its progeny; therefore the predicted BV of progeny (o) is:

$$EBV = \frac{(BV_{sire} + BV_{Dam})}{2}$$

$$Acc = 1/2\sqrt{(Acc_{Sire})^2 + (Acc_{Dam})^2}$$

***Example 4:** Suppose that the estimated BVs for the sire & dam of a heifer are 180 & 70kg respectively for yearling body weight. Given that the accuracy of the proofs are 0.97 for the sire & 0.77 for the dam, predict the BV of the heifer & its accuracy for yearling weight.*

***Solution:** EBV = 125kg* *Acc = 0.619 = 61.9%*

EBV from Multiple Sources - Selection Index

- When records are available from multiple sources, e.g. records on the animal itself, its dam, half-sibs, progeny, etc., it will obviously be most beneficial to use all records to estimate the breeding value.
- It is the **best linear prediction (BLP)** of an individual breeding value. The numerical value obtained for each animal is referred to as the **index (I)** and it is the basis on which animals are ranked for selection.

- Suppose y_1, y_2 and y_3 are phenotypic values for animal **i** and its **sire** and **dam**, the index for this animal using this information would be:

$$I_i = \hat{a}_i = b_i(y_1 - \mu_1) + b_2(y_2 - \mu_2) + b_3(y_3 - \mu_3)$$

- where b_1, b_2 and b_3 are the factors by which each measurement is weighted (**partial regression coefficients**).

- **Disadvantages of selection index (BLP)**

- Records have to be **pre-adjusted** for fixed or environmental factors and these are assumed to be known. But not usually known especially **when no prior data exist** for new subclasses of fixed effect or new environmental factors.

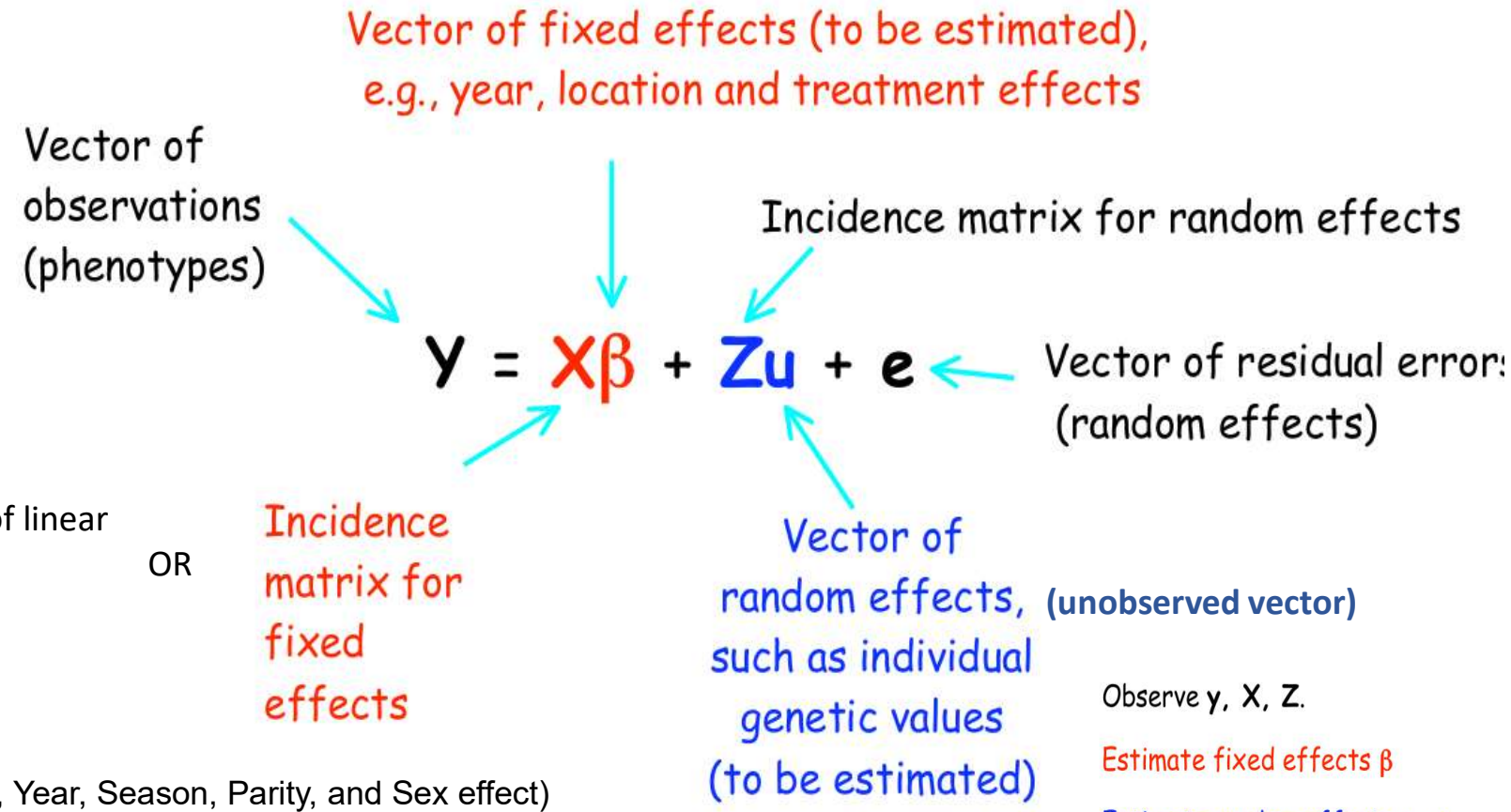
- Solutions to the index equations **require the inverse of the covariance matrix for observations** and this may not be computationally feasible for large data sets.

Best Linear Unbiased Prediction (BLUP) of BV

- **Henderson** (1949) developed a methodology called best linear unbiased prediction (**BLUP**), by which fixed effects and random effects (**BV**) can be simultaneously estimated.

- **BLUP** also called **MME** (Mixed Model Equation) b/c it incorporates both random & fixed effects in the same model. The solution to the MME , thus, gives **BLUE** of fixed effects and **BLUP** of the random effects

BLUP Equation:



coefficient of linear equation

OR

(Herd, Year, Season, Parity, and Sex effect)

Observe y, X, Z.

Estimate fixed effects β

Estimate random effects u, e

BLUP Equation:

- □●□✓□■•◆⌜●□⌟✓▲✓□◭□□✕⌜-▼•✕⌜⌟□▲▨□✓⌜?▲□
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- Because *u* is a random vector rather than a fixed parameter, we talk about **predicting *u* rather than estimating *u***

- We seek a Best Linear Unbiased Predictor (BLUP) for u, which we will denote by u ˆ

- To be a BLUP, we require...

1. u **ˆ to be a linear function of *y*,**

2. *u* **ˆ to be unbiased for *u* so that *E*(*u* ˆ − *u*) = 0, and**

3. Var(*u* ˆ − *u*) to be no "larger" than the Var(*v* − *u*),

In general BLUP

- Uses information on all relatives optimally
- Accounts for fixed effects such as herd, birth type, age
- Accounts for unequal usage of sires in different herds
- Can compare across herd or flocks but need links to exist
- Accounts for culling and selection, non-random mating

  Allows selection across age classes
- Provides an estimate of genetic trend

Derivation of Henderson's Mixed Model Equation (BLUP) gives simplified form

$$\begin{bmatrix} \mathbf{X}'\mathbf{X} & \mathbf{X}'\mathbf{Z} \\ \mathbf{Z}'\mathbf{X} & \mathbf{Z}'\mathbf{Z}+\sigma_e^2\mathbf{G}^{-1} \end{bmatrix}\begin{bmatrix} \mathbf{b} \\ \mathbf{u} \end{bmatrix}=\begin{bmatrix} \mathbf{X}'\mathbf{Y} \\ \mathbf{Y}'\mathbf{Z} \end{bmatrix}$$

Assuming Additivity

$$\mathbf{G}=\mathbf{A}\sigma_a^2$$

$$\begin{bmatrix} \mathbf{X}'\mathbf{X} & \mathbf{X}'\mathbf{Z} \\ \mathbf{Z}'\mathbf{X} & \mathbf{Z}'\mathbf{Z}+\frac{\sigma_e^2}{\sigma_a^2}\mathbf{A}^{-1} \end{bmatrix}\begin{bmatrix} \mathbf{b} \\ \mathbf{u} \end{bmatrix}=\begin{bmatrix} \mathbf{X}'\mathbf{Y} \\ \mathbf{Y}'\mathbf{Z} \end{bmatrix}$$

- **Knowledge of Matrix algebra** is necessary for the understanding of principles of BV estimation using Henderson's Mixed Model Equation (BLUP)

Definition of a Matrix

An n x m (*n-by-m*) matrix is a rectangular array of elements with n rows & m columns. The matrix B, below consists of 2 rows & 3 columns:

$$B = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b23 \end{bmatrix} = \begin{bmatrix} 2 & 4 & 5 \\ 6 & 8 & 9 \end{bmatrix}$$

B is a 2 x 3 matrix with elements b_{ij} (i = 1, 2; j = 1, 2, 3) & sometimes written as B_{2x3}

By convention, matrices in text are printed in bold face.

Matrix Dimension/Order

The dimension (order) of a matrix is given by the number of rows & columns in the matrix (*i.e. the dimensions are r by c*). By convention, rows are always mentioned first. For the matrices below: A is 2 by 2 whereas D is 4 by 3

$$A = \begin{bmatrix} 1 & 3 \\ 7 & 9 \end{bmatrix} \qquad D = \begin{bmatrix} 4 & 2 & 4 \\ 1 & 6 & 0 \\ 3 & 0 & 5 \\ 2 & 3 & 0 \end{bmatrix}$$

The individual elements of a matrix can be referred to by specifying the row & column in which it occurs. Lower case numbers are used to represent individual elements & by convention, elements are printed in italics. For example, individual elements from matrices **A** & **D** above can be referred to as,

$a_{11} = 1$; $a_{21} = 7$; $d_{22} = 6$; $d_{12} = 2$

A matrix called B of order 4 by 4 might look like this:

$$B = \begin{bmatrix} 1.0 & 0.0 & 0.0 & 0.5 \\ 0.0 & 1.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 1.0 & 0.0 \\ 0.5 & 0.0 & 0.0 & 1.9 \end{bmatrix}$$

In the matrix B, for example, $b_{31} = 0$, $b_{22}=1$.

In general, b_{ij} means the element of matrix B in the ith *row* & j^{th} *column*.

Dimensions of a Matrix

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

- 2 Rows
- 3 Columns
- A 2 X 3 Matrix

- 1 X 4

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \end{bmatrix}$$

- 4 X 1

$$A = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

- 2 X 3

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

Definition of a Vector

A matrix consisting of a single column is called a column vector. Thus an n x 1 (***n-by-1***) matrix is called a column vector with n elements:

$$A = \begin{bmatrix} a_1 \\ a_2 \\ \cdot \\ \cdot \\ \cdot \\ a_n \end{bmatrix}$$

A column vector *d* with four elements is:

$$d = \begin{bmatrix} 4 \\ 6 \\ 8 \\ 2 \end{bmatrix}$$

Only one subscript is needed to specify the position of an element

Definition of a Vector

Con't

A matrix consising of a single row of elements is called a row vector. Thus, a row vector is a 1-by-m matrix of numbers. For example, a row vector consisting of 3 elements may be represented as:

b=(2 6 -4)

Only one subscript is needed to specify the position of an element in a row vector:

Examle: **b1 = 2** **b2 = 6** **b3 = -4**

The dimensions of a row vector are (1 by m), where "m" is the number of columns, & the dimensions of a column vector (n by 1), where "n" is the number of rows.

Note: All matrices & vectors will be in bold print to differentiated them from scalar variables. By convention, vectors are printed as lower case bold face letters.

Definition of a scalar matrix

A scalar matrix is a matrix with one row & one column.

Example: 3

Special Matrices

I. Square matrix:

-Is a matrix with an equal number of **rows** & **columns**. Shown below are 2 matrices **A** & **D**:

$$A = \begin{bmatrix} 1 & 3 \\ 7 & 9 \end{bmatrix} \qquad D = \begin{bmatrix} 4 & 2 & 4 \\ 1 & 6 & 0 \\ 3 & 0 & 5 \\ 2 & 3 & 0 \end{bmatrix}$$

-Matrix A above is a square matrix (2 by 2), matrix D is not a square matrix (b/ c it is 4 by 3). Matrix D is a rectangular matrix.

- the **ij** elemnts in a square matrix with **i=j** are called the ***diagonal elements.***

-Other elements of the square matrix are called ***off-diagonal*** or ***non-diagonal*** elements. The diagonal elements in the **A** matrix above are ***1*** & ***9***.

Special Matrices

II. Symmetric Matrix

- Is a square matrix with the elements above the diagonal equal to elements below the diagonal, i.e., $a_{ij} = a_{ji}$

$$A = \begin{bmatrix} 2 & -4 & 0 \\ -4 & 6 & 3 \\ 0 & 3 & 7 \end{bmatrix}$$

For a symmetric matrix, the transpose of the matrix is the original matrix, i.e., $A = A'$.

All symmetric matrices are also square matrices, but all square matrices are not necessarily symmetric matrices!

III. Diagonal Matrix:

A diagonal matrix is a square matrix having zero on all of its off-diagonal elements. Thus, a diagonal matrix is also symmetric matrix. For example:

$$A = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

When all the diagonal elemens of a diagonal matrix are 1, it is called an identity matrix.

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

IV. Triangular Matrix

A triangular matrix is a square matrix in which all elements above/below the diagonal are zeros. For example, the matrix D below is a lower triangular matrix, i.e., all elements above the diagonal are zeros:

$$D = \begin{bmatrix} 4 & 0 & 0 \\ 1 & 3 & 0 \\ 2 & 7 & 9 \end{bmatrix}$$

The matrix E below is an upper-triangular-matrix, i.e., all elements below the diagonal are zeros.

$$E = \begin{bmatrix} 3 & 9 & 1 \\ 0 & 4 & 8 \\ 0 & 0 & 6 \end{bmatrix}$$

BASIC MATRIX OPERATIONS:

1. Matrix Transpose

The transpose of a matrix A, usually denoted as A` or A^T, is the matrix whose ji elements are the ij elements of the original matrix. $A'_{ji} = A_{ij}$.

The columns of A´ are the rows of A & the rows of A´ are the columns of A.

$$Example A = \begin{bmatrix} 3 & 2 \\ 1 & 1 \\ 4 & 0 \end{bmatrix};\quad A^T = \begin{bmatrix} 3 & 1 & 4 \\ 2 & 1 & 0 \end{bmatrix}$$

NB: $A \neq A^T$. But the transpose of a symmetric matrix is equal to the symmetric matrix.

The transpose of a matrix is denoted with a single quote and called prime. For example see **A'** (A prime) & **B'** (B prime) below:

$$\mathbf{A} = \begin{bmatrix} 3 & 1 \\ 4 & 2 \\ 2 & 2 \end{bmatrix}$$

$$\mathbf{A'} = \begin{bmatrix} 3 & 4 & 2 \\ 1 & 2 & 2 \end{bmatrix}$$

$$\mathbf{B} = \begin{bmatrix} 1 & 0 & 0 & .5 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ .5 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{B'} = \begin{bmatrix} 1 & 0 & 0 & .5 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ .5 & 0 & 0 & 1 \end{bmatrix}$$

2. Matrix Addition & Subtraction

Matrices can be added & subtracted if & only if they are of the **same order** (i.e., *they are identical in their number of rows & columns*).

$$A=\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \quad and \quad B=\begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \end{bmatrix}$$

Then:

$$A+B=\begin{bmatrix} (a_{11}+b_{11}) & (a_{12}+b_{12}) & (a_{13}+b_{13}) \\ (a_{21}+b_{21}) & (a_{22}+b_{22}) & (a_{23}+b_{23}) \end{bmatrix}$$

$$A-B=\begin{bmatrix} (a_{11}-b_{11}) & (a_{12}-b_{12}) & (a_{13}-b_{13}) \\ (a_{21}-b_{21}) & (a_{22}-b_{22}) & (a_{23}-b_{23}) \end{bmatrix}$$

Example

$$A=\begin{bmatrix}1 & -2\\ 3 & 4\\ 9 & 0\end{bmatrix}\quad B=\begin{bmatrix}1 & 4\\ 1 & 4\\ -4 & 4\end{bmatrix}\quad A+B=\begin{bmatrix}1+1 & -2+4\\ 3+1 & 4+4\\ 9-4 & 0+4\end{bmatrix}=\begin{bmatrix}2 & 2\\ 4 & 8\\ 5 & 4\end{bmatrix}$$

3. Matrix Multiplication

To do matrix multiplication, **the number of column of the 1st matrix must be equal to the number of rows of the 2nd matrix**. The order of the resulting matrix will be $\mathbf{A.B_{r1xc2}}$.

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \quad and \quad B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{bmatrix}$$

$$A \cdot B$$

$$= \begin{bmatrix} (a_{11}b_{11}+a_{12}b_{21}+a_{13}b_{31}) & (a_{11}b_{12}+a_{12}B_{22}+a_{13}b_{32}) \\ (a_{21}b_{11}+a_{22}b_{21}+a_{23}b_{31}) & (a_{21}b_{12}+a_{22}b_{22}+a_{23}b_{32}) \end{bmatrix}$$

↓ must be equal ↓

(3 x 2) x (2 x 3)

↑ new dimensions ↑

$$A = \begin{bmatrix} 1 & -2 \\ 3 & 4 \\ 9 & 0 \end{bmatrix} \quad X = \begin{bmatrix} 1 & -2 \\ 3 & 4 \end{bmatrix}$$

dimensions for the product of A * X are,

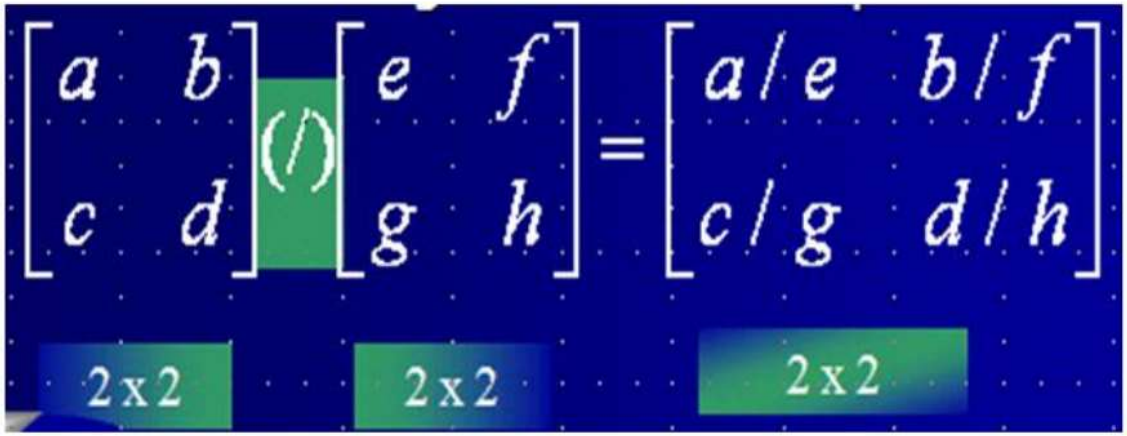
Note that though we can multiply A * X, we could not have done the multiplication the other way (i.e. X * A), since the dimensions would not have matched. That is, matrix multiplication is not necessarily ***commutative!***

Multiplication by a scalar - in this type of multiplication each element of the matrix is simply multiplied, element by element, by the scalar value.

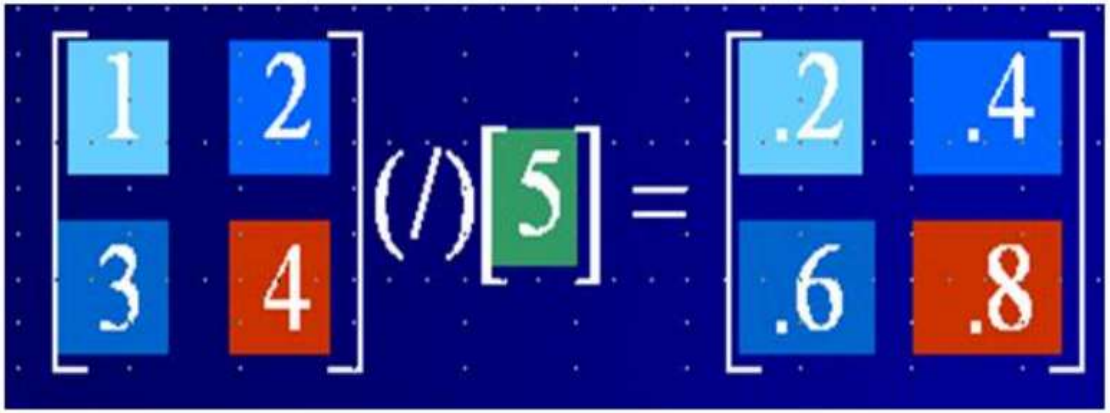
$$A = \begin{bmatrix} 1 & -2 \\ 3 & 4 \\ 9 & 0 \end{bmatrix} \quad B = [7] \qquad A * B = 7 * \begin{bmatrix} 1 & -2 \\ 3 & 4 \\ 9 & 0 \end{bmatrix} = \begin{bmatrix} 7 & -14 \\ 21 & 28 \\ 63 & 0 \end{bmatrix}$$

Division of Matrix Elements

- All matrices must have the same dimensions or the divisor must be a scalar number.
- The division sign is enclosed in parentheses.



Division of Matrix Elements



Matrix Inversion

If A is an n x n matrix (square matrix) having n linearly independent columns (Rank(A)=n) then there exists a matrix A^{-1} such that

$$A.A^{-1} = A^{-1}.A = I_{nxn}$$

An invese matrix is one which, when mltiplied by the original matrix, gives an identity matrix as the product.

Only square matrices can be inverted.

SIMPLE MATRIX INVERSION (2 by 2 matrix only)

The inverse of a (2 by 2) matrix is given by,

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \qquad A^{-1} = \frac{1}{(a \times d) - (b \times c)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

The scalar value resulting from the calculation "(axd) – (bxc)" is called the determinant. The matrix cannot be inverted unless the inverse of the determinant exists (is defined). It will not exist in a case such as the one below since (1÷0) is not defined.

$$A = \begin{bmatrix} 1 & 4 \\ 2 & 8 \end{bmatrix} \qquad \text{then } \frac{1}{\text{Determinant of A}} = \frac{1}{(1 \times 8) - (2 \times 4)} = \frac{1}{0}$$

Matrix Inversion

Example:

$$A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$$

$$A^{-1} = \left(\frac{1}{2\cdot 4 - 3\cdot 1}\right)\cdot \begin{bmatrix} 4 & -3 \\ -1 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{4}{5} & \frac{-3}{5} \\ \frac{-1}{5} & \frac{2}{5} \end{bmatrix}$$

$$\text{Note } A\cdot A^{-1} = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}\cdot \begin{bmatrix} \frac{4}{5} & \frac{-3}{5} \\ \frac{-1}{5} & \frac{2}{5} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Inverse of 3 x3 matrix

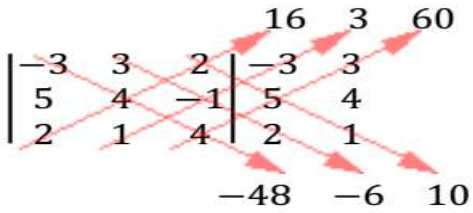
- Find the determinant matrix first

Example 1: Find the determinant of $\begin{vmatrix} -3 & 3 & 2 \\ 5 & 4 & -1 \\ 2 & 1 & 4 \end{vmatrix}$.

Step 1: Rewrite the first two columns of the matrix.

$$\begin{vmatrix} -3 & 3 & 2 \\ 5 & 4 & -1 \\ 2 & 1 & 4 \end{vmatrix} \rightarrow \begin{vmatrix} -3 & 3 & 2 & -3 & 3 \\ 5 & 4 & -1 & 5 & 4 \\ 2 & 1 & 4 & 2 & 1 \end{vmatrix}$$

Step 2: Multiply diagonally downward and diagonally upward.



Step 3: Add the downward numbers together.

$-48 + (-6) + 10 = -44$

Step 4: Add the upward numbers together.

$16 + 3 + 60 = 79$

Step 5: Subtract the upward sum from the downward sum to get the determinant.

$-44 - 79 = -123$

In general, the computation of A^{-1} from A is a tedious task. However, computers are there for performing these calculations.

Several general-purpose computer packages for BLUP evaluations, such as PEST, MTDFREML, ASREML, BREEDPLAN etc. have been written & made available.

BLUP Models:

A) sire model

B) animal model, reduced animal model, animal model with groups

C) maternal model

D) repeatability model, etc.

S I R E M O D E L

- Its application implies that only **sires are evaluated**, using progeny records
- Its main advantage is that the number of equations is reduced compared with an animal model since only sires are evaluated.
- Its **disadvantage** is that the genetic merit of the mate (**dam of progeny) is not accounted for.**

$$y_{ij} = \mu + s_i + e_{ij}$$

- where

y_{ij} the milk yield of the j-th cow of the i-th sire

μ Population average

s_i the random effect of the i-th sire

e_{ij} the random error effect

Example: Sire model of BLUP

Sex of progeny	sire	Sire of sire	Dam of sire	WWG (kg)
M	1	-	-	4.5
F	3	-	-	2.9
F	1	-	-	3.9
M	4	1	-	3.5
M	3	-	-	5.0

Objective:

- to estimate sex effect
- to predict BVs for sires 1, 3 and 4

Given:

$$\sigma_s^2 = 5$$

$$\sigma_e^2 = 55$$

$$\alpha = \frac{\sigma_e^2}{\sigma_s^2} = 11$$

Setting the design matrices and MMEs

- The design matrix X relates to the sex records.
- In the transposed matrix: The first row indicates that the first, fourth and fifth observations are from male calves and the second row shows that the second and third records are from female calves.

Sex of progeny

$$X = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \\ 1 & 0 \\ 1 & 0 \end{bmatrix}$$

$$X' = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} \mathbf{X'X} & \mathbf{X'Z} \\ \mathbf{Z'X} & \mathbf{Z'Z} + \frac{\sigma_e^2}{\sigma_a^2}\mathbf{A}^{-1} \end{bmatrix} \begin{bmatrix} \mathbf{b} \\ \mathbf{u} \end{bmatrix} = \begin{bmatrix} \mathbf{X'Y} \\ \mathbf{Y'Z} \end{bmatrix}$$

Matrix Z

- The design matrix Z relates to the sires records

$$\mathbf{Z}= \begin{bmatrix} \mathbf{1} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{1} & \mathbf{0} \\ \mathbf{1} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{1} \\ \mathbf{0} & \mathbf{1} & \mathbf{0} \end{bmatrix} \text{sires records} \qquad \mathbf{Z}' = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

❖ Sires 1 and 3 have two records each while sire 4 has only one record

Vector of observation (y)

- The vector of observation is y (*5 by 1 vector OR column vector)*

$$y = \begin{bmatrix} 4.5 \\ 2.9 \\ 3.9 \\ 3.5 \\ 3.0 \end{bmatrix}$$

- Having set up the matrices X, Z and y, the matrices X′X, X′Z, Z′X, Z′Z, X′Y & Z′Y in the MME can easily be calculated through matrix multiplication. Thus
- X′X = $\begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$ X′Z = $\begin{bmatrix} \mathbf{1} & \mathbf{1} & \mathbf{1} \\ \mathbf{1} & \mathbf{1} & \mathbf{0} \end{bmatrix}$
- Z′X = Z′X is the transpose of X′Z Z′Z = diagonal matrix $\begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ $\begin{bmatrix} \mathbf{X'X} & \mathbf{X'Z} \\ \mathbf{Z'X} & \mathbf{Z'Z}+\frac{\sigma_e^2}{\sigma_a^2}\mathbf{A}^{-1} \end{bmatrix}\begin{bmatrix} \mathbf{b} \\ \mathbf{u} \end{bmatrix}=\begin{bmatrix} \mathbf{X'Y} \\ \mathbf{Y'Z} \end{bmatrix}$

  i.e $\begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 0 \end{bmatrix}$
- X′Y = $\begin{bmatrix} 13 \\ 6.8 \end{bmatrix}$ Z′Y = $\begin{bmatrix} 8.4 \\ 7.9 \\ 3.5 \end{bmatrix}$
- the LSE are
- * Apart from the fact that sire 4 is the son of sire 1, no other r/nships exist among the three sires:
- A = A^{-1} =
- * The MME obtained after adding $A^{-1}\alpha$ to Z′Z in the LSE are
- * The solution to the MME by direct inversion of the coefficient matrix are

❖ Having set up the matrices X, Z and y, the matrices X′X, X′Z, Z′X, Z′Z, X′Y & Z′Y in the MME can easily be calculated through matrix multiplication for the least square equations.

$$\begin{bmatrix} \mathbf{X'X} & \mathbf{X'Z} \\ \mathbf{Z'X} & \mathbf{Z'Z}+\frac{\sigma_e^2}{\sigma_a^2}\mathbf{A}^{-1} \end{bmatrix}\begin{bmatrix} \mathbf{b} \\ \mathbf{u} \end{bmatrix}=\begin{bmatrix} \mathbf{X'Y} \\ \mathbf{Y'Z} \end{bmatrix}$$

Thus

- X′X = $\begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$ X′Z = $\begin{bmatrix} \mathbf{1} & \mathbf{1} & \mathbf{1} \\ \mathbf{1} & \mathbf{1} & \mathbf{0} \end{bmatrix}$
- Z′X = Z′X is the transpose of X′Z Z′Z = diagonal matrix $\begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

  i.e $\begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 0 \end{bmatrix}$
- X′Y = $\begin{bmatrix} 13 \\ 6.8 \end{bmatrix}$ Z′Y = $\begin{bmatrix} 8.4 \\ 7.9 \\ 3.5 \end{bmatrix}$

For the example data set, the LSE are:

$$\begin{bmatrix} \mathbf{X}'\mathbf{X} & \mathbf{X}'\mathbf{Z} \\ \mathbf{Z}'\mathbf{X} & \mathbf{Z}'\mathbf{Z}+\frac{\sigma_e^2}{\sigma_a^2}\mathbf{A}^{-1} \end{bmatrix} \begin{bmatrix} \mathbf{b} \\ \mathbf{u} \end{bmatrix} = \begin{bmatrix} \mathbf{X}'\mathbf{Y} \\ \mathbf{Y}'\mathbf{Z} \end{bmatrix}$$

$$\begin{bmatrix} 3 & 0 & 1 & 1 & 1 \\ 0 & 2 & 1 & 1 & 0 \\ 1 & 1 & 2 & 1 & 0 \\ 1 & 1 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \hat{b}_1 \\ \hat{b}_2 \\ \hat{s}_1 \\ \hat{s}_3 \\ \hat{s}_4 \end{bmatrix} = \begin{bmatrix} 13.0 \\ 6.8 \\ 8.4 \\ 7.9 \\ 3.5 \end{bmatrix}$$

- The prediction of breeding value requires the inverse of the relationship matrix, A−1.

The relationship matrix can be expressed (Thompson, 1977a) as:

$$A = TDT'$$

where T is a lower triangular matrix and D is a diagonal matrix.

• Say t_{ij} is the coefficient of relationship between animals i and j, can easily be computed applying the following rules.

• For the ith animal:

tij = 1

If both parents (s and d) are known:

t

ij = 0.5(tsj + tdj)

If only one parent (s) is known:

t

ij = 0.5(tsj)

If neither parent is known:

t

ij = 0

Apart from the fact that sire 4 is the son of sire 1, no other relationships exist among the three sires. Therefore A−1 for the three sires is:

$$\mathbf{A}^{-1} = \begin{bmatrix} 1.333 & 0.0 & -0.667 \\ 0.000 & 1.0 & 0.000 \\ -0.667 & 0.0 & 1.333 \end{bmatrix}$$

The MME obtained after adding $\mathbf{A}^{-1}\alpha$ to $\mathbf{Z}'\mathbf{Z}$ in the LSEs are:

$$\begin{bmatrix} \hat{b}_1 \\ \hat{b}_2 \\ \hat{s}_1 \\ \hat{s}_3 \\ \hat{s}_4 \end{bmatrix} \begin{bmatrix} 3.000 & 0.000 & 1.000 & 1.000 & 1.000 \\ 0.000 & 2.000 & 1.000 & 1.000 & 0.000 \\ 1.000 & 1.000 & 16.666 & 0.000 & -7.334 \\ 1.000 & 1.000 & 0.000 & 13.000 & 0.000 \\ 1.000 & 0.000 & -7.334 & 0.000 & 15.666 \end{bmatrix}^{-1} = \begin{bmatrix} 13.00 \\ 6.80 \\ 8.40 \\ 7.90 \\ 3.50 \end{bmatrix}$$

The solutions to the MME by direct inversion of the coefficient matrix are:

Effects	Solutions
Sex	
1	4.336
2	3.382
Sire	
1	0.022
3	0.014
4	–0.043